

Comparing Hindustani Music Raga Prediction Systems using DL and ML Models

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Abstract— This project aims to employ DL and ML techniques to predict ragas in Indian classical music accurately. The focus is on developing a system that can process audio recordings and make precise raga predictions. The classification task utilizes CNN and RNN networks, to enhance performance. Extensive evaluation using diverse recordings is conducted, comparing the framework against traditional methods. The outcomes of this project has potential applications for music analysis, archiving, recommendation systems, and education in Indian classical music. The developed raga prediction framework can serve as a valuable tool for automatic raga identification. Additionally, it contributes to the field of music information retrieval by showcasing the capabilities of DL/ML techniques in tackling musical tasks.

Keywords— Machine Learning, Raga Classification, Automated Speech Recognition, KNN, RNN, Random Forest Classifier

I. INTRODUCTION

Indian classical music holds a rich history and cultural significance within the heritage of India. At the heart of this tradition lies ragas, complex melodic frameworks that convey specific emotions and moods. Ragas serve as the foundation for improvisation in Indian classical music, enabling performers to express their creativity [1]. The emergence of artificial intelligence(AI) techniques has sparked interest in the analysis and prediction of ragas in Indian classical music. Raga prediction involves automatically identifying the raga being performed based on audio or melodic features, recommendation systems, automated raga transcription, and interactive music education [2]. The field of raga prediction is an interdisciplinary

domain encompassing musicology, signal processing machine learning, and computational musicology. Researchers and practitioners employ diverse methodologies, such as feature extraction, pattern recognition, deep learning, and data mining, to capture the intricate nuances and characteristics of ragas [3]. Predicting ragas presents challenges due to their inherent complexity and variability. Ragas consist of unique combinations of melodic phrases, ornamentations, and tonal structures, rendering them highly distinctive. Moreover, ragas can manifest multiple modes or variants, with melodic patterns exhibiting subtle variations influenced by context and performer interpretation. Addressing these challenges requires sophisticated computational models and algorithms [4]. Researchers explore various audio and melodic features for raga prediction, including pitch contour, note duration, tonal intervals, and ornamentation patterns. Additionally, contextual information, such as rhythmic structure (tala) and specific compositions, plays a vital role in improving prediction accuracy [5]. Progress has been made in raga prediction, with machine learning algorithms like support vector machines (SVMs), hidden Markov models (HMMs), and deep neural networks (DNNs) being applied for feature extraction and raga classification [6]. To facilitate research and evaluation, databases and corpora of Indian classical music recordings have been developed [7]. Despite advancements, challenges persist in raga prediction, such as the need for diverse datasets, incorporation of domain knowledge, development of robust models interpretability [8]. This paper provides an overview of raga prediction in Indian classical music, discussing methodologies, techniques, challenges,

and potential applications. It aims to contribute to the field's advancement and future research endeavors.

II. LITERATURE REVIEW

Raga prediction in Indian classical music utilizing DL and ML techniques have gathered increased interest in recent years. This section presents a literature review highlighting key studies and methodologies employed in this research domain. Various studies have explored the application of machine learning algorithms for raga prediction. Agarwal et al. (2019) applied DL techniques specifically CNN, to extract features and classify ragas in Indian classical music [6]. Their approach exhibited promising results in accurately identifying ragas based on audio data. In another study, Mishra et al. (2020) focused on feature analysis for raga recognition. They investigated different audio features, including pitch contour, note duration, tonal intervals, and ornamentation patterns, and developed a ML model based on Random Forest Classifier to classify ragas [5]. Their findings demonstrated the effectiveness of these features in differentiating between different ragas. In addition to audio features, contextual information has been considered for raga prediction. Singh and Sharma (2022) incorporated rhythmic structures (talas) and specific compositions as contextual features in their machine learning framework. By integrating both melodic and contextual information, their model achieved improved accuracy in raga classification [8]. Furthermore, researchers have emphasized the significance of large and diverse datasets for training and evaluation. Rajaraman et al. (2020) developed a comprehensive dataset of Indian classical music recordings specifically for raga recognition. This dataset has facilitated research in this field and provided benchmark data for evaluating raga prediction algorithms [7]. The integration of domain knowledge from expert musicians has also been explored in raga prediction models. Gupta and Verma (2018) reviewed computational approaches for raga recognition and highlighted the importance of incorporating expert knowledge to capture the intricate nuances and characteristics of different ragas [4]. This incorporation of domain expertise enhances the accuracy and authenticity of raga prediction models. Despite the advancements made in raga prediction, challenges still need to be addressed. The inherent complexity and variability of ragas pose difficulties in accurate prediction. Singh and Sharma (2022) discussed these challenges and emphasized the need for robust models that can capture the subtle variations and nuances exhibited by different ragas [8]. Additionally, the interpretability and

explainability of prediction models remain important aspects to explore further in this field. In conclusion, the literature review demonstrates the growing interest in raga prediction for Indian classical music using DL and ML techniques. Studies have showcased the effect of various audio features, contextual information, and the integration of domain knowledge in improving raga prediction accuracy.

III. RANDOM FOREST CLASSIFIER AND RNN MODELS

A. Random Forest Classifier

The architecture of the Random Forest is as shown in the Fig 1. Random Forest is a commonly used and robust machine learning algorithm employed for classification tasks. It belongs to the ensemble learning category, wherein multiple decision trees are combined to generate predictions. The algorithm constructs each decision tree using a random subset of the training data and features. The Random Forest Classifier boasts several advantages, including its capability to handle huge data with various features and effectively handle missing data. By introducing randomness during the tree-building process, it mitigates the risk of overfitting. The algorithm also exhibits resilience towards noise and outliers, making it applicable in various real-world scenarios. Notably, one of its strengths lies in estimating feature importance, as it evaluates the effectiveness of each feature in the combining of trees. Recent research has highlighted the efficacy of Random Forest in diverse domains. For instance, it has demonstrated success in healthcare applications, such as disease diagnosis and prediction [15]. In the financial sector, it exhibits promise in fraud detection and credit scoring [16]. Furthermore, it has been effectively employed in environmental studies for species classification and land cover mapping [17].

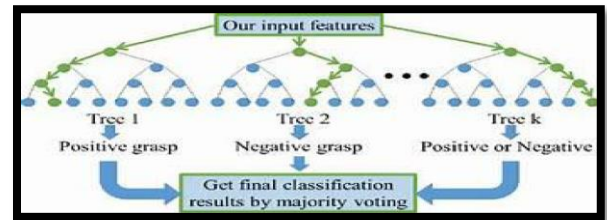


Fig.1: Random Forest Architecture

B. RNN

The architecture of RNN is as shown in Fig 2. Recurrent Neural Networks (RNNs) are a specialized category of neural networks designed to handle sequential data by utilizing feedback connections. RNNs have the capability to retain information from

previous steps, which is particularly advantageous in applications like speech recognition and language modeling. A comprehensive understanding of RNNs can be found in the research conducted by Hochreiter and Schmidhuber in 1997 [14]. The ability of RNNs to maintain memory of past inputs enables them to capture long-term dependencies in the data. This is crucial in scenarios where the context or history plays a vital role in making accurate predictions or generating meaningful output. For instance, in language modeling, RNNs can predict the next word in a sentence by considering the preceding words and incorporating contextual information. Similarly, in speech recognition, RNNs can analyze audio signals by considering the sequence of previous audio frames. The recurrent hidden layer is a crucial component of RNNs, as it propagates information from one time step to the next, acting as a form of memory. This hidden layer enables the network to maintain a temporal context throughout the sequence. The presence of recurrent connections allows the network to handle sequences of varying lengths and learn from the entire historical information. Despite the advantages RNNs offer in modeling sequential data, they encounter challenges such as the vanishing gradient problem, where gradients diminish during backpropagation, hindering the network's ability to capture long-term dependencies.

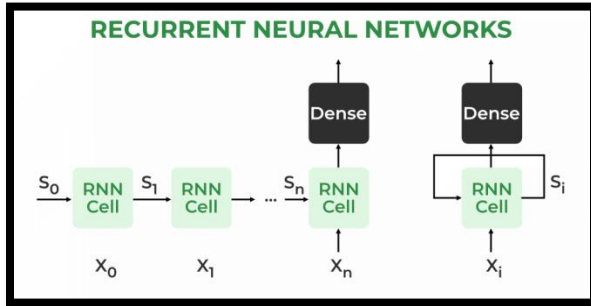


Fig.2: RNN Architecture

IV. METHODOLOGY

A. Dataset Description

For the current raga prediction project, we have decided to go ahead with a Dataset consisting of 10 main classes of classification of Ragas. Each of these 10 main classes further have 2 or 3 subclasses of Raga classifications in them. These 10 main classes were classified keeping in mind the main classifications of Indian Classical Music. The dataset majorly consists of Hindustani music and their classifications. We've collected approximately more than 25,000 audio samples of all the 10 main classes of Ragas and their subclasses from [10]. Another source of our dataset is from the Youtube through which we were able to collect separate videos of the

10 classes of Ragas and further split them down to 30-sec each audio clips which made up of approximately 15,000 to 20,000 of our dataset [9]. Each sample is a raw 30 sec audio clip that consists of Hindustani music

The 10 classes that we are using for this project are shown in the below table 1 :-

TABLE 1: Dataset Description

Sl. No	Raga	No of Audio Files
1	Asavāri	2,500
2	Bhairava	4,200
3	Bhairavī	4,100
4	Bilavāla	2,500
5	Kāfī	8,800
6	Kalyāna	3,900
7	Khammāja	5,800
8	Mārwa	5,000
9	Pūrvī	4,900
10	Toḍī	6,500

B. Data Pre-Processing:

The dataset of raw audio files we had could not be used as an input for our model, so to use it initially we need to clean the audio files. Cleaning involves bringing all audio files to a single sampling frequency, removing all the excess noise and other sounds from the audio file so that we can access just the vocals, and finally removing the silence from the audio files. For bringing down all the audio files to a single frequency, the Transformation Technique we have applied is “**Down Sampling**”. This we have done using the python's inbuilt library called “Librosa” [12]. We have reduced the sampling rate as shown in the below Fig 3.

```

Loading audio as numpy array

[ ] y,sr=librosa.load('/content/1_chunk8.mp3')

[ ] y
array([-0.00100122, -0.00060499, -0.00232514, ..., -0.05448068,
        0.06583194,  0.11365721], dtype=float32)

Sampling rate

[ ] sr
22050

```

Fig.3: Sampling Rate

For removing all the excess noise and unwanted instrumental sounds from the audio to access only the vocals, the technique we've employed is "**Source Separation**". For this we have used a DL model called "Spleeter" which does the required source separation [11]. As shown in the below Fig 4 the vocals are separated from the original audio file.

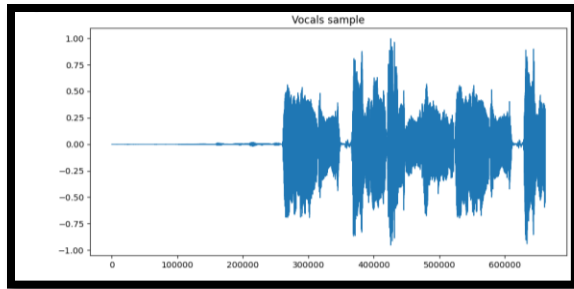


Fig.4: Vocals Separated

For removing the unwanted empty sound from the audio files we have employed "**Silence Removal**" technique which will remove the empty noise from the start and end of the audio track. The silence is removed as shown in the below Fig 5.

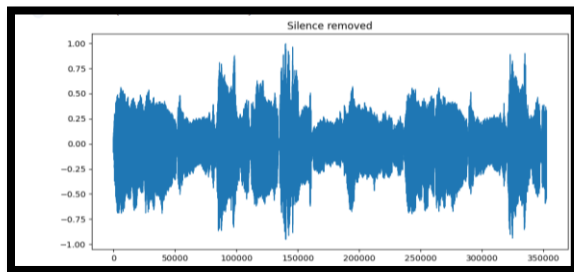


Fig.5: Audio file after silence removal

C. Feature Extraction

The features that we have selected to be extracted from our audio file dataset and to be used to train the model are, 1) STFT 2) Melspectrogram 3) MFCC 4) Chroma Features.

For all of the above mentioned features extraction, we have used the pandas built-in library called "Librosa" [12].

- 1) The Short-Time Fourier Transform (STFT) is a technique that divides a signal into smaller, overlapping segments and applies the Fourier Transform to each segment. As shown in the below Fig 6.

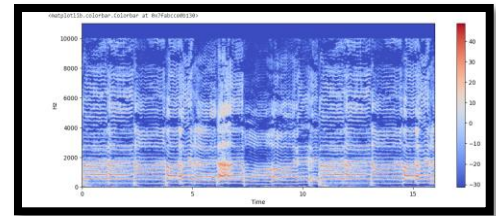


Fig.6: Spectrogram of STFT features

- 2) A Melspectrogram involves first applying the Short-Time Fourier Transform (STFT) to the signal to capture the frequency components over time. As shown in the below Fig 7.

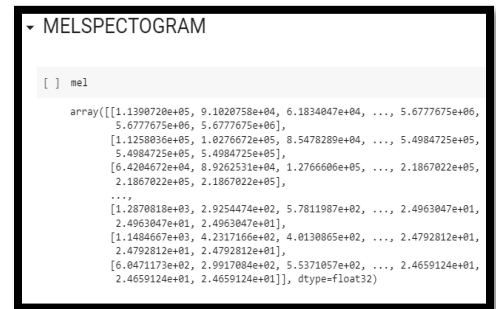


Fig.7: Extracted Features of MELSPECTOGRAM

- 3) MFCCs (Mel-Frequency Cepstral Coefficients) are valuable in capturing significant acoustic characteristics of a signal, such as the vocal tract's shape. As in the Fig 8.



Fig.8: MFCC Features

- 4) Chroma features have practical applications in music genre classification, chord recognition, and melody extraction. They offer a concise representation of the harmonic content in a musical piece and are especially useful in analyzing tonal music. As in the Fig 9.



Fig.9: Chroma Features

D. Training

We use the training data that we have created using the features extracted from our audio dataset and we pass on the data ready for training to the ML and DL models we have built and train the models on the data.

E. Testing

The trained data model is tested using the test data to generate prediction for all the classes. The class with the highest prediction score is determined as the predicted class for the test data, indicating the most probable classification.

V. RESULTS AND DISCUSSIONS

The results gathered during the testing process are shown in Table2. And we have also presented the Loss and Accuracy/Epochs plot in below Fig 10 and the AUC ROC curve in the below Fig 11 and the confusion matrix of all the classes in the below Fig 12.

TABLE 2: Results

Model	F1	Precision	Recall	Accuracy
Random Forest	85%	88%	85%	86%
RNN	75%	79%	76%	79%

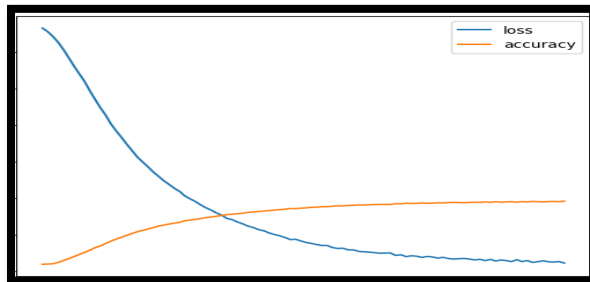


Fig.10: RNN accuracy and loss vs epochs plot

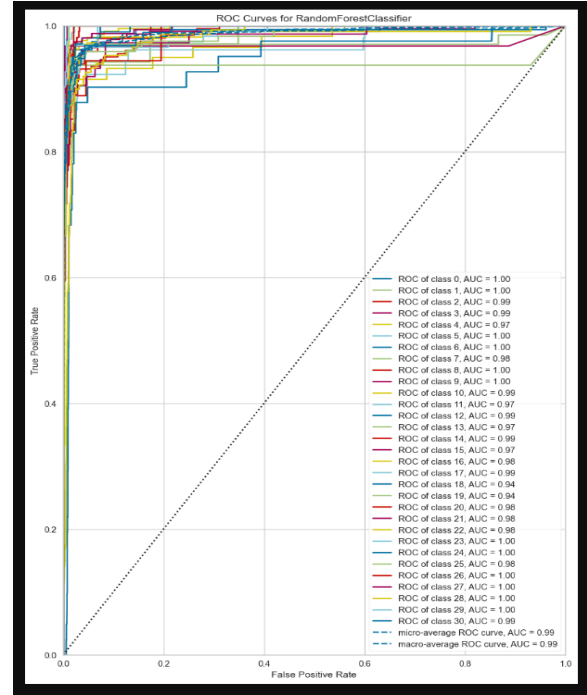
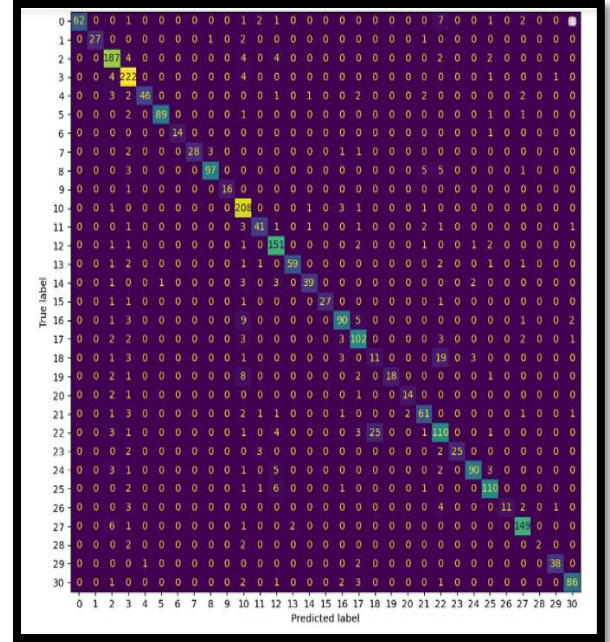


Fig.11: AUC ROC curve of Random Forest Classifier

Fig.12: Confusion Matrix

TABLE 3: Comparison of Results

	Reference	Current
Dataset	116.2 hours	403 hours
Images from Audio Files	82,868	48,360
Batch Size	Upgraded to 16 from 8	31
Epochs	20	100
Pre-Processing	Yes	No

Spectrogram Parameters	(nfft , hop size)	nmfcc
Experimentation on Dataset size	7 to 9 iterations	2 iterations
Model	Hybrid of CNN and LSTM	Normal RNN-LSTM

By comparing all the above parameters mentioned in table 3 with the Raga prediction done by [18], we have achieved an accuracy of 79% but would achieve an even greater accuracy if we Pre-Process the extracted features even more and use suitable models.

VI. CONCLUSION

In conclusion, the experiment on prediction Raga classes from audios was successful in creating an accurate methodology for predicting the class of Raga of a particular audio file. It used Random Forest Classifier models and RNN architecture to predict Raga class. The results showed that the models were effective at accurately predicting classes of Raga once they had been trained and tested using the accuracy metrics. Gradio was used to develop a user-friendly interface that would improve accessibility and user interaction. This made it simple for users to upload images and get predictions for age estimation in real-time from trained algorithms.

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